Exponential Functions & Growth Models

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential growth and decay basics**

Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.

$f(x) = 5(1.2)^x$: growth, since the base $1.2 > 1$ (growth rate 20% per unit x). $g(x) = 5(0.8)^x$: decay, since the base $0.8 < 1$ (decay rate 20% per unit x).

2 **[Foundational]** • **Target: Exponential growth and decay basics**

A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.

The form $2^{t/3}$ means the population doubles every 3 hours. After 9 hours (3 doubling periods): $P(9) = 1000(2)^3 = 1000(8) = 8000$ bacteria.

3 **[Foundational]** • **Target: Visualizing exponential growth**

The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.

a Identify which curve represents growth and which represents decay. $f(x) = 2^x$ is the growth curve (increasing, base > 1). $g(x) = 3(0.5)^x$ is the decay curve (decreasing, base < 1).

b State the horizontal asymptote shared by both curves. Both curves approach the horizontal asymptote $y = 0$ (neither ever reaches or crosses it).

(a) $f(x) = 2^x$ is the growth curve (increasing, base > 1). $g(x) = 3(0.5)^x$ is the decay curve (decreasing, base < 1).; (b) Both curves approach the horizontal asymptote $y = 0$ (neither ever reaches or crosses it).

4 **[Foundational]** • **Target: Finding an exponential model from data**

An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

$2420 = 2000(1 + r)^2$: $(1 + r)^2 = 1.21$, so $1 + r = \sqrt{1.21} = 1.1$, giving $r = 0.10 = 10\%$.

5 **[Foundational]** • **Target: Finding an exponential model from data**

A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80\,b^t$, to four decimal places.

$45 = 80\,b^6$: $b^6 = \frac{45}{80} = 0.5625$. $b = (0.5625)^{1/6} \approx 0.9128$.

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6 [Foundational] • Target: Continuous growth and the number e

A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.

$A(20) = 500e^{0.03(20)} = 500e^{0.6} \approx 500(1.8221) \approx 911.$

7 [Foundational] • Target: Continuous growth and the number e

Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r.

The discrete model compounds growth once per period (e.g. once a year), while the continuous model compounds growth constantly at every instant. They're related by $e^k = 1 + r$, i.e. $k = \ln(1 + r)$ — the continuous rate k is slightly less than the discrete rate r for positive growth, since continuous compounding reaches the same total growth with a slightly lower nominal rate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Explain why, for large enough x, any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.

Exponential functions grow by repeated multiplication, so their rate of growth itself keeps growing exponentially, while a polynomial's growth rate eventually stabilizes into a fixed power-law pattern. Even a very slowly-growing exponential like 1.01^x will eventually overtake x^{10}, because exponential growth compounds indefinitely while polynomial growth does not — it just takes an enormous value of x (in this case, extremely large, since 1.01^x grows very slowly at first) for the crossover to occur.

9 [Intermediate] • Target: Applications: half-life and doubling time

A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.

$\frac{A(20)}{A_0} = \left(\frac{1}{2}\right)^{20/8} = \left(\frac{1}{2}\right)^{2.5} = 2^{-2.5} \approx 0.1768$, so about 17.68% remains.

10 [Intermediate] • Target: Applications: half-life and doubling time

A city's population doubles every 15 years and currently stands at 40{,}000. Write an exponential model $P(t)$ and predict the population after 45 years.

$P(t) = 40000(2)^{t/15}$. After 45 years ($t/15 = 3$ doubling periods):
$P(45) = 40000(2)^3 = 40000(8) = 320{,}000.$

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11 [Intermediate] • Target: Transformations of exponential functions

Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.

$g(x) = 2^{x-3} + 4$ shifts $f(x) = 2^x$ right 3 units and up 4 units. The horizontal asymptote shifts from $y = 0$ to $y = 4$.

12 [Intermediate] • Target: Transformations of exponential functions

Find the horizontal asymptote and y-intercept of $h(x) = -3(2)^x + 6$.

The negative coefficient reflects the graph over the horizontal asymptote line; as $x \to \infty$, $-3(2)^x \to -\infty$, so $h(x) \to -\infty$, while as $x \to -\infty$, $-3(2)^x \to 0$, giving horizontal asymptote $y = 6$. y-intercept: $h(0) = -3(1) + 6 = 3$, so $(0, 3)$.

13 [Intermediate] • Target: Compound interest and financial models

A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0\left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.

$A(3) = 5000(1.01)^{12} \approx 5000(1.1268) \approx 5634$ dollars.

14 [Intermediate] • Target: Compound interest and financial models

Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.

More frequent compounding applies the (smaller) periodic interest rate more often, and interest starts earning interest sooner and more often within each year. This pushes the effective annual growth rate slightly above the nominal rate, so daily compounding yields a marginally larger balance than quarterly compounding at the same stated annual rate.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t = 15$: $0.80M_0 = M_0 e^{-15k} \Rightarrow e^{-15k} = 0.80$. Take the natural logarithm of both sides: $-15k = \ln(0.80) \Rightarrow k = -\frac{\ln(0.80)}{15} = \frac{\ln(1.25)}{15} \approx 0.01488$ day^{-1}. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}} = 0.5$: $t_{1/2} = \frac{\ln 2}{k} = \frac{15\ln 2}{-\ln(0.80)} \approx \frac{15(0.6931)}{0.2231} \approx 46.6$ days.

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16 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A Ferris wheel has a diameter of 40 meters and its lowest point is 4 meters above the ground. Passengers board at the lowest point at $t = 0$, and the wheel completes one full revolution every 8 minutes. Write a cosine model $h(t)$ for a passenger's height at time t, and find the times during the first revolution when $h(t) = 34$ meters.

Step 1: Determine sinusoidal parameters: Radius (amplitude) $= 20$ m; midline $D = 4 + 20 = 24$ m; angular frequency $B = \frac{2\pi}{8} = \frac{\pi}{4}$. Starting at the minimum gives a negative cosine: $\mathbf{h(t) = -20\cos\left(\frac{\pi}{4}t\right) + 24}$. Step 2: Solve $h(t) = 34$ on $[0, 8]$: $-20\cos\left(\frac{\pi}{4}t\right) + 24 = 34 \Rightarrow \cos\left(\frac{\pi}{4}t\right) = -\frac{1}{2}$. .

17 [Challenging] • Target: Exponential growth and decay basics • Cross-Lesson: 1.02 Rational Functions & Asymptotes

The half-life of Carbon-14 is 5730 years. (a) Write $A(t)$. (b) How much of 200 g remains after 11 460 years? (c) When is only 10% left?

(a) Exponential Decay Function: $A(t) = 200\left(\frac{1}{2}\right)^{t/5730}$. (b) Mass Remaining at $t = 11{,}460$ years: Notice that $\frac{11{,}460}{5730} = 2$ half-lives: $A(11{,}460) = 200\left(\frac{1}{2}\right)^2 = 200\left(\frac{1}{4}\right) = \mathbf{50}$ g. (c) Time to Reach 10% of Initial Mass: Set $\frac{A(t)}{200} = 0.10$.

18 [Challenging] • Target: Visualizing exponential growth • Cross-Lesson: 1.03 Rates of Change & Concavity

[Unit 1 · Polynomial and Rational Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Finding an exponential model from data • Cross-Lesson: 2.02 Logarithmic Functions & Properties

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Continuous growth and the number e • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Level 1 · FoundationsFind the exact value of $\tan\left(\frac{11\pi}{6}\right)$ and $\sec\left(\frac{11\pi}{6}\right)$ using reference angles and unit circle coordinates.

Quadrant & Reference Angle: $\frac{11\pi}{6}$ lies in Quadrant IV because $\frac{3\pi}{2}$\theta_{\text{ref}} = 2\pi - \frac{11\pi}{6} = \frac{\pi}{6}.x and \(y $\cos\left(\frac{11\pi}{6}\right) = +\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$, $\sin\left(\frac{11\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}$. Calculate Ratios: $\tan\left(\frac{11\pi}{6}\right) = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$. .

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exponential growth and decay basics

A population is modeled by $P(t) = 900(3)^{t/3}$, where t is in years. Find $P(6)$.

a. 2700

b. 5400

c. 8100 ✓

d. 24300

Since $t/3 = 2$: $P(6) = 900(3)^2 = 900(9) = 8100$.

22 [Foundational] • Target: Visualizing exponential growth

A substance has a half-life of 6 days. What fraction of the original amount remains after 18 days?

a. $\frac{1}{4}$

b. $\frac{1}{16}$

c. $\frac{3}{8}$

d. $\frac{1}{8}$ ✓

Since $t/6 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

23 [Foundational] • Target: Finding an exponential model from data

An investment grows from 300 dollars to 507 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

a. $\frac{3}{10}$ ✓

b. $\frac{2}{5}$

c. $-\frac{3}{10}$

d. $\frac{13}{10}$

$507 = 300(1 + r)^2$: $(1 + r)^2 = \frac{507}{300}$, so $1 + r = \frac{13}{10}$, giving $r = \frac{3}{10}$.

24 [Foundational] • Target: Continuous growth and the number e

A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{20})^{nt}$, where $nt = 15$. If 2000 dollars is deposited, find the balance to the nearest dollar.

a. 4008

b. 4258

c. 2000

d. 4158 ✓

$A = 2000(1 + \frac{1}{20})^{15} \approx 2000(2.0789) \approx 4158$ dollars.

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25 [Foundational] • Target: Comparing growth rates: exponential vs. polynomial

Which is an exponential decay?

a. $y = 3^x$

b. $y = 0.4^x$ ✓

c. $y = e^x$

d. $y = 10^x$

Applying Unit 2 · Exponential and Logarithmic Functions principles, Base < 1 ⇒ decay. Therefore, the verified correct option is $y = 0.4^x$.

26 [Foundational] • Target: Applications: half-life and doubling time

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$ ✓

Applying Exponential & Logarithmic Functions principles, Standard half-life model. Therefore, the verified correct option is $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$.

27 [Foundational] • Target: Transformations of exponential functions

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$ ✓

b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

Applying Exponential & Logarithmic Functions principles, As $n \to \infty$: $A = Pe^{rt}$. Therefore, the verified correct option is $A = Pe^{rt}$.

28 [Foundational] • Target: Compound interest and financial models

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8

b. 2

c. 4 ✓

d. 10

Applying Exponential & Logarithmic Functions principles, $2^{10/5} = 2^2 = 4$. Therefore, the verified correct option is 4.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Exponential growth and decay basics**

For $f(x) = 2(0.8)^x$, the function represents:

a. exponential growth of 20% per unit x

b. no growth or decay (constant function)

c. linear change with rate 0.8 per unit x

d. **exponential decay of 20% per unit x** ✓

Since $b = 0.8$ is between 0 and 1, this is exponential decay; the percent change per unit x is x

30 **[Intermediate]** • **Target: Visualizing exponential growth**

Solve for x (round to 3 decimal places): $5^x = 15$.

a. 3.000

b. 2.183

c. **1.683** ✓

d. 1.183

Take $\log$ of both sides: $x\log 5 = \log 15$, so $x = \frac{\log 15}{\log 5} \approx 1.683$.

31 **[Intermediate]** • **Target: Finding an exponential model from data**

By the Remainder Theorem, what is the remainder when $f(x) = 1x^3 + 5x^2 - 4x + 4$ is divided by $(x + 4)$?

a. 34

b. -36

c. 72

d. **36** ✓

By the Remainder Theorem, the remainder equals $f(-4) = 1(-4)^3 + 5(-4)^2 - 4(-4) + 4 = 36$.

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32 [Intermediate] • Target: Continuous growth and the number e

A bacteria culture triples every 4 hours. Starting with 200 bacteria, the population after t hours is

a. $200 \cdot 3^{t/4}$ ✓

b. $200 \cdot 3^{4t}$

c. $200 + 3t$

d. $600 \cdot t$

P(t) = P■ · b^(t/period) = 200 · 3^(t/4).

33 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Remainder when $p(x) = x^3 - 2x + 1$ is divided by $(x - 2)$:

a. 3

b. **5** ✓

c. 7

d. 9

$p(2) = 8 - 4 + 1 = 5$ by Remainder Theorem.

34 [Intermediate] • Target: Applications: half-life and doubling time

$A = P(1 + r/n)^{nt}$. Find A (to the nearest cent) when x2000, r=0.04, n=2, t=2.

a. 2154.86

b. 2174.86

c. **\2164.86** ✓

d. 2163.20

x2164.86.

35 [Intermediate] • Target: Transformations of exponential functions

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. **0.0434** ✓

We are given the exponential growth model $P(t) = P_0 e^{kt}$. We know that the initial population $P_0 = 500$ and that after 10 years, $P(10) = 1200$. We can substitute these values into the model: $1200 = 500e^{k \cdot 10}$ Divide both sides by 500: $\frac{1200}{500} = e^{10k}$ $2.4 = e^{10k}$ To solve for k, we take the natural logarithm of both sides: $\ln(2.4) = \ln(e^{10k})$ $\ln(2.4) = 10k$ Now, divide by 10: $k = \frac{\ln(2.4)}{10}$ Using a calculator, $\ln(2.4) \approx 0.8754687$. So.

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36 [Intermediate] • Target: Compound interest and financial models

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347** ✓

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

37 [Intermediate] • Target: Exponential growth and decay basics

For $A = A_0e^{-kt}$, find the half-life $t_{1/2}$ in terms of k.

a. $\frac{k}{\ln 2}$

b. $\frac{\ln 2}{k}$ ✓

c. $\frac{1}{k}$

d. e^k

$1/2 = e^{-kt_{1/2}} \Rightarrow \ln(1/2) = -kt_{1/2} \Rightarrow t_{1/2} = \ln 2/k$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing exponential growth • Cross-Lesson: 1.03 Rates of Change & Concavity

[Transformation Trial 1] Identify the matrix representing a counterclockwise rotation by 360° in $\mathbb{R}^2$.

a. $(-1, 0; 0, 1)$

b. $(0, 1; 1, 0)$

c. $(1, 0; 0, 1)$ ✓

d. $(1, -1; 1, 1)$

The 2D rotation matrix is $(\cos\theta, -\sin\theta; \sin\theta, \cos\theta)$. For $\theta = 360^\circ$, this evaluates to $(1, 0; 0, 1)$.

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39 [Challenging] • Target: Finding an exponential model from data • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Pythagorean identity: $1 + \tan^2\theta =$

a. $\csc^2\theta$

b. $\sec^2\theta$ ✓

c. $\cot^2\theta$

d. $\cos^2\theta$

Applying Unit 3 · Trigonometric and Polar Functions principles, $\sec^2\theta = 1 + \tan^2\theta$ Therefore, the verified correct option is $\sec^2\theta$.

40 [Challenging] • Target: Continuous growth and the number e • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Convert the rectangular point $(-1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

a. $r = 1.414,\ \theta = 225°$

b. $r = 2.414,\ \theta = 135°$

c. $r = 1.414,\ \theta = 315°$

d. $r = 1.414,\ \theta = 135°$ ✓

$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (1)^2} \approx 1.414$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 135°$.

41 [Challenging] • Target: Comparing growth rates: exponential vs. polynomial • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Eliminate the parameter to find y as a function of x: $x = 1t - 2$, $y = -2t^2$.

a. $y = -1(x + 2)^2$

b. $y = -2(x + 2)^2$ ✓

c. $y = 2(x + 2)^2$

d. $y = -2(x - 2)^2$

From $x = 1t - 2$, $t = \frac{x+2}{1}$. Substitute into $y = -2t^2$: $y = -2\left(\frac{x+2}{1}\right)^2 = -2(x + 2)^2$.

42 [Challenging] • Target: Applications: half-life and doubling time • Cross-Lesson: 4.01 Inverse & Composite Functions

Write a sinusoidal function with amplitude 5, period 4π, midline $y = -3$, using $\cos$.

a. $f(x) = 5\cos(1/2x) - 3$ ✓

b. $f(x) = 5\cos(1/2x) - 2$

c. $f(x) = 6\cos(1/2x) - 3$

d. $f(x) = 5\cos(1x) - 3$

Amplitude $= A = 5$; period $= 2\pi/B = 4\pi \Rightarrow B = 1/2$; midline $D = -3$. So $f(x) = 5\cos(1/2x) - 3$.

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43 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Identify the polar curve $r = 5 + 3\cos\theta$.

a. a cardioid

b. a circle

c. **a dimpled limaçon ✓**

d. a rose with 3 petals

The equation $r = 5 + 3\cos\theta$ has the standard polar form matching: a dimpled limaçon.

44 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 4.03 Parametric Equations & Modeling

Find the component form of vector $\mathbf{u}$ directed from $(9, 12)$ to $(20, -2)$.

a. $\langle 12, -14 \rangle$

b. $\langle -11, 14 \rangle$

c. $\langle 11, -14 \rangle$ ✓

d. $\langle 11, 14 \rangle$

$\mathbf{u} = \langle (20) - (9), (-2) - (12) \rangle = \langle 11, -14 \rangle$.

45 [Challenging] • Target: Exponential growth and decay basics • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Does the point $(4, 0)$ satisfy the implicitly defined curve $x^2 + y^2 = 16$?

a. No, the point does not lie on the curve

b. The equation does not define a curve

c. **Yes, the point lies on the curve ✓**

d. Only if r is negative

Substitute: $(4)^2 + (0)^2 = 16$, compare to 16: equal, so it is on the curve.

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